

EduSaaS Full Database Object Catalog

Source: Edu_saas_DB_Scripts.sql plus approved backend-owned AI-support additions for 5 new AI tables, recommendation extension columns, AI indexes, and updated_at triggers.

Purpose: one consolidated database-object catalog covering schema, enums, tables, indexes, foreign keys, functions, triggers, and explicitly noted missing/not-present object types from the source SQL.

1. Object Inventory Summary

Object Type	Count	Source / Notes
Schema statements	1	Extracted from source SQL
Extensions	0	None found in source SQL
Enums	4	Extracted from source SQL
CREATE TABLE statements	41	Extracted from source SQL
AI support CREATE TABLE statements	5	Added from approved backend-owned AI requirements
Total tables covered	46	41 baseline + 5 AI support
CREATE INDEX statements	76	Extracted from source SQL
Foreign key ALTER statements	56	Extracted from source SQL
Functions in source SQL	0	None found in source SQL
Triggers in source SQL	0	None found in source SQL
Views in source SQL	0	None found in source SQL
Sequences in source SQL	0	None explicit; SERIAL columns create sequences automatically in PostgreSQL
Seed INSERT statements	0	None found in source SQL
Comments	0	None found in source SQL
Grants	0	None found in source SQL

2. Baseline Table Inventory

#	Table Name
1	education.users
2	education.roles
3	education.permissions
4	education.role_permissions
5	education.profiles
6	education.assessments
7	education.gap_reports
8	education.courses
9	education.enrollments
10	education.jobs
11	education.applications
12	education.notifications
13	education.subscriptions
14	education.reports
15	education.teachers

© 2026 Vayvora Technologies Pvt. Ltd. All Rights Reserved. VAYVORA® (Trademark Applied) — TM Application No.: 7874082

16	education.lessons
17	education.quizzes
18	education.assignments
19	education.progress
20	education.certificates
21	education.achievements
22	education.tasks
23	education.recommendations
24	education.announcements
25	education.auth_otps
26	education.refresh_tokens
27	education.domain_roles
28	education.skills
29	education.domain_required_skills
30	education.difficulty_levels
31	education.questions
32	education.quiz_sessions
33	education.student_answers
34	education.student_skill_results
35	education.quiz_state
36	education.community_bookmarks
37	education.community_comments
38	education.community_post_tags
39	education.community_posts
40	education.community_reactions
41	education.community_tags

3. AI Support Table Inventory

#	Table Name
42	education.activity_logs
43	education.login_history
44	education.course_skills
45	education.course_prerequisites
46	education.course_ratings

4. Schema, Extensions, and Enum DDL

4.1 Schema DDL

CREATE SCHEMA IF NOT EXISTS "education";

4.2 Extension DDL

No CREATE EXTENSION statements were found in the provided source SQL. If gen_random_uuid() is used in the target database, pgcrypto should be enabled separately if not already available.

4.3 Enum DDL

Enum 1: Object 1

```
CREATE TYPE "education"."community_post_status" AS ENUM ('Draft', 'Published', 'Archived', 'Flagged');
```

Enum 2: Object 2

```
CREATE TYPE "education"."community_post_type" AS ENUM ('Discussion', 'Question', 'Course', 'Announcement', 'Job', 'Internship', 'Project', 'Achievement', 'Event');
```

Enum 3: Object 3

```
CREATE TYPE "education"."community_reaction_type" AS ENUM ('Like', 'Celebrate', 'Support', 'Insightful');
```

Enum 4: Object 4

```
CREATE TYPE "education"."community_visibility" AS ENUM ('Public', 'Students', 'Educators', 'Employers', 'Admins');
```

5. Baseline CREATE TABLE DDL from Source SQL

Baseline Table 1: users

```
CREATE TABLE "education"."users" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "name" VARCHAR(255) NOT NULL,  
  "email" VARCHAR(255) NOT NULL,  
  "role_id" UUID NOT NULL,  
  "password_hash" TEXT NOT NULL,  
  "status" VARCHAR(50) NOT NULL DEFAULT 'active',  
  "last_login" TIMESTAMP(6),  
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  "domain_role_id" UUID,  
  
  CONSTRAINT "users_pkey" PRIMARY KEY ("id")  
);
```

Baseline Table 2: roles

```
CREATE TABLE "education"."roles" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "name" VARCHAR(100) NOT NULL,  
  "description" TEXT,  
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "roles_pkey" PRIMARY KEY ("id")  
);
```

Baseline Table 3: permissions

```
CREATE TABLE "education"."permissions" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "name" VARCHAR(150) NOT NULL,  
  "description" TEXT,  
  "category" VARCHAR(100),  
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "permissions_pkey" PRIMARY KEY ("id")  
);
```

```
);
```

Baseline Table 4: role_permissions

```
CREATE TABLE "education"."role_permissions" (  
  "role_id" UUID NOT NULL,  
  "permission_id" UUID NOT NULL,  
  "assigned_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "role_permissions_pkey" PRIMARY KEY ("role_id","permission_id")  
);
```

Baseline Table 5: profiles

```
CREATE TABLE "education"."profiles" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "user_id" UUID NOT NULL,  
  "career_goal" TEXT,  
  "institution" TEXT,  
  "company" TEXT,  
  "preferences" JSONB,  
  "initial_assessment_completed" BOOLEAN NOT NULL DEFAULT false,  
  
  CONSTRAINT "profiles_pkey" PRIMARY KEY ("id")  
);
```

Baseline Table 6: assessments

```
CREATE TABLE "education"."assessments" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "user_id" UUID NOT NULL,  
  "type" VARCHAR(100) NOT NULL,  
  "score" DOUBLE PRECISION NOT NULL,  
  "answers" JSONB,  
  "date_taken" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  "career_goal" TEXT,  
  "is_initial" BOOLEAN NOT NULL DEFAULT false,  
  "total_marks" INTEGER,  
  "obtained_marks" INTEGER,  
  "completed" BOOLEAN NOT NULL DEFAULT false,  
  "started_at" TIMESTAMP(6),  
  "completed_at" TIMESTAMP(6),  
  
  CONSTRAINT "assessments_pkey" PRIMARY KEY ("id")  
);
```

Baseline Table 7: gap_reports

```
CREATE TABLE "education"."gap_reports" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "user_id" UUID NOT NULL,  
  "readiness_score" DOUBLE PRECISION NOT NULL,  
  "missing_skills" TEXT[],  
  "recommendations" JSONB,  
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  "updated_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "gap_reports_pkey" PRIMARY KEY ("id")  
);
```

Baseline Table 8: courses

```
CREATE TABLE "education"."courses" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "title" VARCHAR(255) NOT NULL,  
  "description" TEXT,  
  "provider" VARCHAR(255),  
  "category" VARCHAR(255),  
  "difficulty" VARCHAR(50),  
  "status" VARCHAR(50) NOT NULL DEFAULT 'active',  
  "educator_id" UUID,  
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "courses_pkey" PRIMARY KEY ("id")  
);
```

Baseline Table 9: enrollments

```
CREATE TABLE "education"."enrollments" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "user_id" UUID NOT NULL,  
  "course_id" UUID NOT NULL,  
  "status" VARCHAR(50) NOT NULL DEFAULT 'active',  
  "completion_percentage" DOUBLE PRECISION NOT NULL DEFAULT 0,  
  "enrolled_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "enrollments_pkey" PRIMARY KEY ("id")  
);
```

Baseline Table 10: jobs

```
CREATE TABLE "education"."jobs" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "employer_id" UUID NOT NULL,  
  "title" VARCHAR(255) NOT NULL,  
  "description" TEXT,  
  "requirements" TEXT,  
  "required_skills" TEXT[],  
  "status" VARCHAR(50) NOT NULL DEFAULT 'open',  
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "jobs_pkey" PRIMARY KEY ("id")  
);
```

Baseline Table 11: applications

```
CREATE TABLE "education"."applications" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "job_id" UUID NOT NULL,  
  "student_id" UUID NOT NULL,  
  "status" VARCHAR(50) NOT NULL DEFAULT 'submitted',  
  "skill_match" DOUBLE PRECISION,  
  "applied_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "applications_pkey" PRIMARY KEY ("id")  
);
```

Baseline Table 12: notifications

```
CREATE TABLE "education"."notifications" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "user_id" UUID NOT NULL,
```

```
"type" VARCHAR(100) NOT NULL,  
"message" TEXT NOT NULL,  
"read_status" BOOLEAN NOT NULL DEFAULT false,  
"created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
CONSTRAINT "notifications_pkey" PRIMARY KEY ("id")  
);
```

Baseline Table 13: subscriptions

```
CREATE TABLE "education"."subscriptions" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "user_id" UUID NOT NULL,  
  "plan_type" VARCHAR(100) NOT NULL,  
  "start_date" TIMESTAMP(6) NOT NULL,  
  "end_date" TIMESTAMP(6) NOT NULL,  
  
  CONSTRAINT "subscriptions_pkey" PRIMARY KEY ("id")  
);
```

Baseline Table 14: reports

```
CREATE TABLE "education"."reports" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "title" VARCHAR(255) NOT NULL,  
  "type" VARCHAR(100) NOT NULL,  
  "generated_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  "exported_at" TIMESTAMP(6),  
  "format" VARCHAR(50),  
  "payload" JSONB,  
  
  CONSTRAINT "reports_pkey" PRIMARY KEY ("id")  
);
```

Baseline Table 15: settings

```
CREATE TABLE "education"."settings" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "scope" VARCHAR(100) NOT NULL DEFAULT 'system',  
  "key" VARCHAR(255) NOT NULL,  
  "value" JSONB NOT NULL,  
  
  CONSTRAINT "settings_pkey" PRIMARY KEY ("id")  
);
```

Baseline Table 16: lessons

```
CREATE TABLE "education"."lessons" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "course_id" UUID NOT NULL,  
  "title" VARCHAR(255) NOT NULL,  
  "video_url" TEXT,  
  "duration" INTEGER,  
  "order_index" INTEGER NOT NULL DEFAULT 0,  
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "lessons_pkey" PRIMARY KEY ("id")  
);
```

Baseline Table 17: quizzes

```
CREATE TABLE "education"."quizzes" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "lesson_id" UUID NOT NULL,  
  "questions" JSONB NOT NULL,  
  "passing_score" INTEGER NOT NULL DEFAULT 60,  
  
  CONSTRAINT "quizzes_pkey" PRIMARY KEY ("id")  
);
```

Baseline Table 18: assignments

```
CREATE TABLE "education"."assignments" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "lesson_id" UUID NOT NULL,  
  "instructions" TEXT,  
  "submission_link" TEXT,  
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "assignments_pkey" PRIMARY KEY ("id")  
);
```

Baseline Table 19: progress

```
CREATE TABLE "education"."progress" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "user_id" UUID NOT NULL,  
  "lesson_id" UUID NOT NULL,  
  "watched_duration" INTEGER NOT NULL DEFAULT 0,  
  "quiz_score" INTEGER,  
  "assignment_status" VARCHAR(50) NOT NULL DEFAULT 'pending',  
  "completion_flag" BOOLEAN NOT NULL DEFAULT false,  
  "updated_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  "completed_at" TIMESTAMP(6),  
  
  CONSTRAINT "progress_pkey" PRIMARY KEY ("id")  
);
```

Baseline Table 20: certificates

```
CREATE TABLE "education"."certificates" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "user_id" UUID NOT NULL,  
  "course_id" UUID NOT NULL,  
  "certificate_code" VARCHAR(255) NOT NULL,  
  "issued_date" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "certificates_pkey" PRIMARY KEY ("id")  
);
```

Baseline Table 21: achievements

```
CREATE TABLE "education"."achievements" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "user_id" UUID NOT NULL,  
  "badge_name" VARCHAR(255) NOT NULL,  
  "milestone" TEXT,  
  "certificate_id" UUID,  
  "earned_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,
```

```
CONSTRAINT "achievements_pkey" PRIMARY KEY ("id")
);
```

Baseline Table 22: tasks

```
CREATE TABLE "education"."tasks" (
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),
  "user_id" UUID NOT NULL,
  "course_id" UUID,
  "title" VARCHAR(255) NOT NULL,
  "due_date" TIMESTAMP(6),
  "status" VARCHAR(50) NOT NULL DEFAULT 'pending',
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,
  "completed_at" TIMESTAMP(3),

  CONSTRAINT "tasks_pkey" PRIMARY KEY ("id")
);
```

Baseline Table 23: recommendations

```
CREATE TABLE "education"."recommendations" (
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),
  "user_id" UUID NOT NULL,
  "course_id" UUID NOT NULL,
  "reason" TEXT,
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT "recommendations_pkey" PRIMARY KEY ("id")
);
```

Baseline Table 24: announcements

```
CREATE TABLE "education"."announcements" (
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),
  "educator_id" UUID NOT NULL,
  "title" VARCHAR(255) NOT NULL,
  "message" TEXT NOT NULL,
  "audience" VARCHAR(100) NOT NULL DEFAULT 'all',
  "scheduled_at" TIMESTAMP(6),
  "attachment" TEXT,
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT "announcements_pkey" PRIMARY KEY ("id")
);
```

Baseline Table 25: auth_otps

```
CREATE TABLE "education"."auth_otps" (
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),
  "user_id" UUID NOT NULL,
  "otp_hash" TEXT NOT NULL,
  "expires_at" TIMESTAMP(6) NOT NULL,
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,
  "verified_at" TIMESTAMP(6),

  CONSTRAINT "auth_otps_pkey" PRIMARY KEY ("id")
);
```

Baseline Table 26: refresh_tokens

```
CREATE TABLE "education"."refresh_tokens" (
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),
```

```

"user_id" UUID NOT NULL,
"token_hash" TEXT NOT NULL,
"expires_at" TIMESTAMP(6) NOT NULL,
"created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,

CONSTRAINT "refresh_tokens_pkey" PRIMARY KEY ("id")
);

```

Baseline Table 27: domain_roles

```

CREATE TABLE "education"."domain_roles" (
  "domain_role_id" UUID NOT NULL DEFAULT gen_random_uuid(),
  "domain_name" VARCHAR(100) NOT NULL,
  "category" VARCHAR(50),
  "created_at" TIMESTAMP(6) DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT "domain_roles_pkey" PRIMARY KEY ("domain_role_id")
);

```

Baseline Table 28: skills

```

CREATE TABLE "education"."skills" (
  "skill_id" SERIAL NOT NULL,
  "skill_name" VARCHAR(100) NOT NULL,
  "category" VARCHAR(50),
  "description" TEXT,
  "created_at" TIMESTAMP(6) DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT "skills_pkey" PRIMARY KEY ("skill_id")
);

```

Baseline Table 29: domain_required_skills

```

CREATE TABLE "education"."domain_required_skills" (
  "domain_required_skill_id" SERIAL NOT NULL,
  "domain_role_id" UUID NOT NULL,
  "skill_id" INTEGER NOT NULL,
  "required_level" SMALLINT NOT NULL,

  CONSTRAINT "domain_required_skills_pkey" PRIMARY KEY ("domain_required_skill_id")
);

```

Baseline Table 30: difficulty_levels

```

CREATE TABLE "education"."difficulty_levels" (
  "difficulty_id" SERIAL NOT NULL,
  "difficulty_name" VARCHAR(20) NOT NULL,
  "difficulty_order" INTEGER NOT NULL,

  CONSTRAINT "difficulty_levels_pkey" PRIMARY KEY ("difficulty_id")
);

```

Baseline Table 31: questions

```

CREATE TABLE "education"."questions" (
  "question_id" SERIAL NOT NULL,
  "skill_id" INTEGER NOT NULL,
  "difficulty_id" INTEGER NOT NULL,
  "question_text" TEXT NOT NULL,
  "option_a" TEXT NOT NULL,
  "option_b" TEXT NOT NULL,

```

```

"option_c" TEXT NOT NULL,
"option_d" TEXT NOT NULL,
"correct_option" CHAR(1) NOT NULL,
"marks" INTEGER DEFAULT 1,
"is_active" BOOLEAN DEFAULT true,
"created_at" TIMESTAMP(6) DEFAULT CURRENT_TIMESTAMP,

CONSTRAINT "questions_pkey" PRIMARY KEY ("question_id")
);

```

Baseline Table 32: quiz_sessions

```

CREATE TABLE "education"."quiz_sessions" (
  "session_id" SERIAL NOT NULL,
  "user_id" UUID NOT NULL,
  "domain_role_id" UUID NOT NULL,
  "start_time" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,
  "end_time" TIMESTAMP(6),
  "status" VARCHAR(20) NOT NULL DEFAULT 'In Progress',
  "total_questions" INTEGER NOT NULL DEFAULT 50,
  "questions_answered" INTEGER NOT NULL DEFAULT 0,
  "current_skill_id" INTEGER,

  CONSTRAINT "quiz_sessions_pkey" PRIMARY KEY ("session_id")
);

```

Baseline Table 33: student_answers

```

CREATE TABLE "education"."student_answers" (
  "answer_id" SERIAL NOT NULL,
  "session_id" INTEGER NOT NULL,
  "skill_id" INTEGER NOT NULL,
  "question_id" INTEGER NOT NULL,
  "difficulty_id" INTEGER NOT NULL,
  "selected_option" CHAR(1) NOT NULL,
  "correct_option" CHAR(1) NOT NULL,
  "is_correct" BOOLEAN NOT NULL,
  "marks_awarded" INTEGER NOT NULL DEFAULT 0,
  "answered_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT "student_answers_pkey" PRIMARY KEY ("answer_id")
);

```

Baseline Table 34: student_skill_results

```

CREATE TABLE "education"."student_skill_results" (
  "result_id" SERIAL NOT NULL,
  "session_id" INTEGER NOT NULL,
  "skill_id" INTEGER NOT NULL,
  "obtained_score" INTEGER NOT NULL,
  "maximum_score" INTEGER NOT NULL,
  "percentage" DECIMAL(5,2) NOT NULL,
  "skill_level" SMALLINT NOT NULL,
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT "student_skill_results_pkey" PRIMARY KEY ("result_id")
);

```

Baseline Table 35: quiz_state

```

CREATE TABLE "education"."quiz_state" (
  "session_id" INTEGER NOT NULL,

```

```

"skill_id" INTEGER NOT NULL,
"current_difficulty" INTEGER DEFAULT 1,
"correct_streak" INTEGER DEFAULT 0,
"wrong_streak" INTEGER DEFAULT 0,
"questions_answered" INTEGER DEFAULT 0,
"obtained_score" INTEGER DEFAULT 0,
"maximum_score" INTEGER DEFAULT 0,
"state" JSONB,

CONSTRAINT "quiz_state_pkey" PRIMARY KEY ("session_id","skill_id")
);

```

Baseline Table 36: community_bookmarks

```

CREATE TABLE "education"."community_bookmarks" (
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),
  "user_id" UUID NOT NULL,
  "post_id" UUID NOT NULL,
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT "community_bookmarks_pkey" PRIMARY KEY ("id")
);

```

Baseline Table 37: community_comments

```

CREATE TABLE "education"."community_comments" (
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),
  "post_id" UUID NOT NULL,
  "author_id" UUID NOT NULL,
  "parent_id" UUID,
  "content" TEXT NOT NULL,
  "reactions_count" INTEGER NOT NULL DEFAULT 0,
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,
  "updated_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,
  "deleted_at" TIMESTAMP(6),

  CONSTRAINT "community_comments_pkey" PRIMARY KEY ("id")
);

```

Baseline Table 38: community_post_tags

```

CREATE TABLE "education"."community_post_tags" (
  "post_id" UUID NOT NULL,
  "tag_id" UUID NOT NULL,
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT "pk_community_post_tags" PRIMARY KEY ("post_id","tag_id")
);

```

Baseline Table 39: community_posts

```

CREATE TABLE "education"."community_posts" (
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),
  "author_id" UUID NOT NULL,
  "title" VARCHAR(255) NOT NULL,
  "content" TEXT NOT NULL,
  "post_type" "education"."community_post_type" NOT NULL,
  "status" "education"."community_post_status" NOT NULL DEFAULT 'Published',
  "visibility" "education"."community_visibility" NOT NULL DEFAULT 'Public',
  "media_url" TEXT,
  "metadata" JSONB,
  "comments_count" INTEGER NOT NULL DEFAULT 0,

```

```

"reactions_count" INTEGER NOT NULL DEFAULT 0,
"created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,
"updated_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,
"deleted_at" TIMESTAMP(6),

CONSTRAINT "community_posts_pkey" PRIMARY KEY ("id")
);

```

Baseline Table 40: community_reactions

```

CREATE TABLE "education"."community_reactions" (
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),
  "user_id" UUID NOT NULL,
  "post_id" UUID,
  "comment_id" UUID,
  "reaction_type" "education"."community_reaction_type" NOT NULL DEFAULT 'Like',
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT "community_reactions_pkey" PRIMARY KEY ("id")
);

```

Baseline Table 41: community_tags

```

CREATE TABLE "education"."community_tags" (
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),
  "name" VARCHAR(50) NOT NULL,
  "slug" VARCHAR(50) NOT NULL,
  "usage_count" INTEGER NOT NULL DEFAULT 0,
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT "community_tags_pkey" PRIMARY KEY ("id")
);

```

6. Additional AI Support CREATE TABLE DDL

AI Table 1: education.activity_logs

```

CREATE TABLE IF NOT EXISTS education.activity_logs (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID NOT NULL,
  sessions_last_30_days INTEGER NOT NULL,
  avg_session_minutes DOUBLE PRECISION,
  videos_watched INTEGER,
  assignments_attempted INTEGER,
  discussion_interactions INTEGER,
  login_count INTEGER,
  device_count INTEGER,
  ip_changes INTEGER,
  last_activity TIMESTAMP WITHOUT TIME ZONE,
  suspicious_activity_score NUMERIC,
  created_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT fk_activity_logs_user
    FOREIGN KEY (user_id) REFERENCES education.users(id)
    ON DELETE CASCADE
);

```

AI Table 2: education.login_history

```
CREATE TABLE IF NOT EXISTS education.login_history (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id UUID NOT NULL,  
  logins_last_30_days INTEGER,  
  days_since_last_login INTEGER,  
  created_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP,  
  updated_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP,  
  CONSTRAINT fk_login_history_user  
    FOREIGN KEY (user_id) REFERENCES education.users(id)  
    ON DELETE CASCADE  
);
```

AI Table 3: education.course_skills

```
CREATE TABLE IF NOT EXISTS education.course_skills (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  course_id UUID NOT NULL,  
  skill_id INTEGER NOT NULL,  
  skill_level INTEGER,  
  created_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP,  
  updated_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP,  
  CONSTRAINT fk_course_skills_course  
    FOREIGN KEY (course_id) REFERENCES education.courses(id)  
    ON DELETE CASCADE,  
  CONSTRAINT fk_course_skills_skill  
    FOREIGN KEY (skill_id) REFERENCES education.skills(skill_id)  
    ON DELETE CASCADE,  
  CONSTRAINT uq_course_skills_course_skill  
    UNIQUE (course_id, skill_id)  
);
```

AI Table 4: education.course_prerequisites

```
CREATE TABLE IF NOT EXISTS education.course_prerequisites (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  course_id UUID NOT NULL,  
  prerequisite_course_id UUID NOT NULL,  
  created_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP,  
  updated_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP,  
  CONSTRAINT fk_course_prerequisites_course  
    FOREIGN KEY (course_id) REFERENCES education.courses(id)  
    ON DELETE CASCADE,  
  CONSTRAINT fk_course_prerequisites_prerequisite  
    FOREIGN KEY (prerequisite_course_id) REFERENCES education.courses(id)  
    ON DELETE CASCADE,  
  CONSTRAINT uq_course_prerequisite  
    UNIQUE (course_id, prerequisite_course_id),  
  CONSTRAINT chk_course_prerequisite_not_same  
    CHECK (course_id <> prerequisite_course_id)  
);
```

AI Table 5: education.course_ratings

```
CREATE TABLE IF NOT EXISTS education.course_ratings (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id UUID NOT NULL,  
  course_id UUID NOT NULL,  
  rating DOUBLE PRECISION NOT NULL,  
  created_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP,  
  updated_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP,  
  CONSTRAINT fk_course_ratings_user  
    FOREIGN KEY (user_id) REFERENCES education.users(id)  
    ON DELETE CASCADE,
```

```

CONSTRAINT fk_course_ratings_course
    FOREIGN KEY (course_id) REFERENCES education.courses(id)
    ON DELETE CASCADE,
CONSTRAINT chk_course_rating
    CHECK (rating >= 1 AND rating <= 5),
CONSTRAINT uq_course_rating_user_course
    UNIQUE (user_id, course_id)
);

```

7. Recommendation Table AI Extension DDL

```

ALTER TABLE education.recommendations
    ADD COLUMN IF NOT EXISTS predicted_rating DOUBLE PRECISION,
    ADD COLUMN IF NOT EXISTS similarity_score DOUBLE PRECISION,
    ADD COLUMN IF NOT EXISTS confidence_score DOUBLE PRECISION,
    ADD COLUMN IF NOT EXISTS prerequisite_completed BOOLEAN,
    ADD COLUMN IF NOT EXISTS rank INTEGER,
    ADD COLUMN IF NOT EXISTS updated_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP;

```

```

ALTER TABLE education.recommendations
    ADD CONSTRAINT chk_recommendations_predicted_rating
        CHECK (predicted_rating IS NULL OR (predicted_rating >= 1 AND predicted_rating <= 5));

```

```

ALTER TABLE education.recommendations
    ADD CONSTRAINT chk_recommendations_similarity_score
        CHECK (similarity_score IS NULL OR (similarity_score >= 0 AND similarity_score <= 1));

```

```

ALTER TABLE education.recommendations
    ADD CONSTRAINT chk_recommendations_confidence_score
        CHECK (confidence_score IS NULL OR (confidence_score >= 0 AND confidence_score <= 1));

```

```

ALTER TABLE education.recommendations
    ADD CONSTRAINT chk_recommendations_rank
        CHECK (rank IS NULL OR rank > 0);

```

8. Baseline Index DDL from Source SQL

#	Index Name
1	users_email_key
2	idx_users_role
3	users_domain_role_id_idx
4	roles_name_key
5	permissions_name_key
6	profiles_user_id_key
7	idx_assessment_user
8	idx_gapreport_user
9	idx_courses_educator
10	idx_enrollment_course
11	idx_enrollment_user
12	uq_enrollment
13	idx_jobs_employer
14	idx_application_job
15	idx_application_student
16	uq_application
17	idx_notification_user

18	idx_subscription_user
19	uq_settings
20	idx_lessons_course
21	idx_quizzes_lesson
22	idx_assignments_lesson
23	idx_progress_lesson
24	idx_progress_user
25	uq_progress
26	certificates_certificate_code_key
27	idx_certificates_course
28	idx_certificates_user
29	idx_achievements_certificate
30	idx_achievements_user
31	idx_tasks_course
32	idx_tasks_user
33	idx_recommendations_course
34	idx_recommendations_user
35	idx_announcements_educator
36	uq_auth_otps_user_id
37	idx_refresh_tokens_expires
38	idx_refresh_tokens_user
39	domain_roles_domain_name_key
40	skills_skill_name_key
41	domain_required_skills_domain_role_id_skill_id_key
42	difficulty_levels_difficulty_name_key
43	difficulty_levels_difficulty_order_key
44	idx_quiz_sessions_current_skill
45	idx_quiz_sessions_domain_role
46	idx_quiz_sessions_status
47	idx_quiz_sessions_user
48	idx_student_answers_difficulty
49	idx_student_answers_question
50	idx_student_answers_session
51	idx_student_answers_skill
52	idx_student_skill_results_session
53	idx_student_skill_results_skill
54	idx_bookmark_post
55	idx_bookmark_user
56	uq_bookmark_user_post
57	idx_community_comments_author
58	idx_community_comments_created
59	idx_community_comments_deleted
60	idx_community_comments_parent
61	idx_community_comments_post
62	idx_post_tags_post
63	idx_post_tags_tag
64	idx_community_posts_author
65	idx_community_posts_created_at
66	idx_community_posts_deleted_at
67	idx_community_posts_post_type
68	idx_community_posts_status
69	idx_community_posts_visibility
70	idx_reaction_comment

71	idx_reaction_post
72	idx_reaction_user
73	uq_community_tag_name
74	uq_community_tag_slug
75	idx_community_tags_slug
76	idx_community_tags_usage

Index 1: users_email_key

CREATE UNIQUE INDEX "users_email_key" ON "education"."users"("email");

Index 2: idx_users_role

CREATE INDEX "idx_users_role" ON "education"."users"("role_id");

Index 3: users_domain_role_id_idx

CREATE INDEX "users_domain_role_id_idx" ON "education"."users"("domain_role_id");

Index 4: roles_name_key

CREATE UNIQUE INDEX "roles_name_key" ON "education"."roles"("name");

Index 5: permissions_name_key

CREATE UNIQUE INDEX "permissions_name_key" ON "education"."permissions"("name");

Index 6: profiles_user_id_key

CREATE UNIQUE INDEX "profiles_user_id_key" ON "education"."profiles"("user_id");

Index 7: idx_assessment_user

CREATE INDEX "idx_assessment_user" ON "education"."assessments"("user_id");

Index 8: idx_gapreport_user

CREATE INDEX "idx_gapreport_user" ON "education"."gap_reports"("user_id");

Index 9: idx_courses_educator

CREATE INDEX "idx_courses_educator" ON "education"."courses"("educator_id");

Index 10: idx_enrollment_course

CREATE INDEX "idx_enrollment_course" ON "education"."enrollments"("course_id");

Index 11: idx_enrollment_user

CREATE INDEX "idx_enrollment_user" ON "education"."enrollments"("user_id");

Index 12: uq_enrollment

CREATE UNIQUE INDEX "uq_enrollment" ON "education"."enrollments"("user_id", "course_id");

Index 13: idx_jobs_employer

```
CREATE INDEX "idx_jobs_employer" ON "education"."jobs"("employer_id");
```

Index 14: idx_application_job

```
CREATE INDEX "idx_application_job" ON "education"."applications"("job_id");
```

Index 15: idx_application_student

```
CREATE INDEX "idx_application_student" ON "education"."applications"("student_id");
```

Index 16: uq_application

```
CREATE UNIQUE INDEX "uq_application" ON "education"."applications"("job_id", "student_id");
```

Index 17: idx_notification_user

```
CREATE INDEX "idx_notification_user" ON "education"."notifications"("user_id");
```

Index 18: idx_subscription_user

```
CREATE INDEX "idx_subscription_user" ON "education"."subscriptions"("user_id");
```

Index 19: uq_settings

```
CREATE UNIQUE INDEX "uq_settings" ON "education"."settings"("scope", "key");
```

Index 20: idx_lessons_course

```
CREATE INDEX "idx_lessons_course" ON "education"."lessons"("course_id");
```

Index 21: idx_quizzes_lesson

```
CREATE INDEX "idx_quizzes_lesson" ON "education"."quizzes"("lesson_id");
```

Index 22: idx_assignments_lesson

```
CREATE INDEX "idx_assignments_lesson" ON "education"."assignments"("lesson_id");
```

Index 23: idx_progress_lesson

```
CREATE INDEX "idx_progress_lesson" ON "education"."progress"("lesson_id");
```

Index 24: idx_progress_user

```
CREATE INDEX "idx_progress_user" ON "education"."progress"("user_id");
```

Index 25: uq_progress

```
CREATE UNIQUE INDEX "uq_progress" ON "education"."progress"("user_id", "lesson_id");
```

Index 26: certificates_certificate_code_key

```
CREATE UNIQUE INDEX "certificates_certificate_code_key" ON "education"."certificates"("certificate_code");
```

Index 27: idx_certificates_course

```
CREATE INDEX "idx_certificates_course" ON "education"."certificates"("course_id");
```

Index 28: idx_certificates_user

```
CREATE INDEX "idx_certificates_user" ON "education"."certificates"("user_id");
```

Index 29: idx_achievements_certificate

```
CREATE INDEX "idx_achievements_certificate" ON "education"."achievements"("certificate_id");
```

Index 30: idx_achievements_user

```
CREATE INDEX "idx_achievements_user" ON "education"."achievements"("user_id");
```

Index 31: idx_tasks_course

```
CREATE INDEX "idx_tasks_course" ON "education"."tasks"("course_id");
```

Index 32: idx_tasks_user

```
CREATE INDEX "idx_tasks_user" ON "education"."tasks"("user_id");
```

Index 33: idx_recommendations_course

```
CREATE INDEX "idx_recommendations_course" ON "education"."recommendations"("course_id");
```

Index 34: idx_recommendations_user

```
CREATE INDEX "idx_recommendations_user" ON "education"."recommendations"("user_id");
```

Index 35: idx_announcements_educator

```
CREATE INDEX "idx_announcements_educator" ON "education"."announcements"("educator_id");
```

Index 36: uq_auth_otps_user_id

```
CREATE UNIQUE INDEX "uq_auth_otps_user_id" ON "education"."auth_otps"("user_id");
```

Index 37: idx_refresh_tokens_expires

```
CREATE INDEX "idx_refresh_tokens_expires" ON "education"."refresh_tokens"("expires_at");
```

Index 38: idx_refresh_tokens_user

```
CREATE INDEX "idx_refresh_tokens_user" ON "education"."refresh_tokens"("user_id");
```

Index 39: domain_roles_domain_name_key

```
CREATE UNIQUE INDEX "domain_roles_domain_name_key" ON "education"."domain_roles"("domain_name");
```

Index 40: skills_skill_name_key

```
CREATE UNIQUE INDEX "skills_skill_name_key" ON "education"."skills"("skill_name");
```

Index 41: domain_required_skills_domain_role_id_skill_id_key

```
CREATE UNIQUE INDEX "domain_required_skills_domain_role_id_skill_id_key" ON "education"."domain_required_skills"("domain_role_id", "skill_id");
```

Index 42: difficulty_levels_difficulty_name_key

```
CREATE UNIQUE INDEX "difficulty_levels_difficulty_name_key" ON "education"."difficulty_levels"("difficulty_name");
```

Index 43: difficulty_levels_difficulty_order_key

```
CREATE UNIQUE INDEX "difficulty_levels_difficulty_order_key" ON "education"."difficulty_levels"("difficulty_order");
```

Index 44: idx_quiz_sessions_current_skill

```
CREATE INDEX "idx_quiz_sessions_current_skill" ON "education"."quiz_sessions"("current_skill_id");
```

Index 45: idx_quiz_sessions_domain_role

```
CREATE INDEX "idx_quiz_sessions_domain_role" ON "education"."quiz_sessions"("domain_role_id");
```

Index 46: idx_quiz_sessions_status

```
CREATE INDEX "idx_quiz_sessions_status" ON "education"."quiz_sessions"("status");
```

Index 47: idx_quiz_sessions_user

```
CREATE INDEX "idx_quiz_sessions_user" ON "education"."quiz_sessions"("user_id");
```

Index 48: idx_student_answers_difficulty

```
CREATE INDEX "idx_student_answers_difficulty" ON "education"."student_answers"("difficulty_id");
```

Index 49: idx_student_answers_question

```
CREATE INDEX "idx_student_answers_question" ON "education"."student_answers"("question_id");
```

Index 50: idx_student_answers_session

```
CREATE INDEX "idx_student_answers_session" ON "education"."student_answers"("session_id");
```

Index 51: idx_student_answers_skill

```
CREATE INDEX "idx_student_answers_skill" ON "education"."student_answers"("skill_id");
```

Index 52: idx_student_skill_results_session

```
CREATE INDEX "idx_student_skill_results_session" ON "education"."student_skill_results"("session_id");
```

Index 53: idx_student_skill_results_skill

```
CREATE INDEX "idx_student_skill_results_skill" ON "education"."student_skill_results"("skill_id");
```

Index 54: idx_bookmark_post

```
CREATE INDEX "idx_bookmark_post" ON "education"."community_bookmarks"("post_id");
```

Index 55: idx_bookmark_user

```
CREATE INDEX "idx_bookmark_user" ON "education"."community_bookmarks"("user_id");
```

Index 56: uq_bookmark_user_post

```
CREATE UNIQUE INDEX "uq_bookmark_user_post" ON "education"."community_bookmarks"("user_id", "post_id");
```

Index 57: idx_community_comments_author

```
CREATE INDEX "idx_community_comments_author" ON "education"."community_comments"("author_id");
```

Index 58: idx_community_comments_created

```
CREATE INDEX "idx_community_comments_created" ON "education"."community_comments"("created_at" DESC);
```

Index 59: idx_community_comments_deleted

```
CREATE INDEX "idx_community_comments_deleted" ON "education"."community_comments"("deleted_at");
```

Index 60: idx_community_comments_parent

```
CREATE INDEX "idx_community_comments_parent" ON "education"."community_comments"("parent_id");
```

Index 61: idx_community_comments_post

```
CREATE INDEX "idx_community_comments_post" ON "education"."community_comments"("post_id");
```

Index 62: idx_post_tags_post

```
CREATE INDEX "idx_post_tags_post" ON "education"."community_post_tags"("post_id");
```

Index 63: idx_post_tags_tag

```
CREATE INDEX "idx_post_tags_tag" ON "education"."community_post_tags"("tag_id");
```

Index 64: idx_community_posts_author

```
CREATE INDEX "idx_community_posts_author" ON "education"."community_posts"("author_id");
```

Index 65: idx_community_posts_created_at

```
CREATE INDEX "idx_community_posts_created_at" ON "education"."community_posts"("created_at" DESC);
```

Index 66: idx_community_posts_deleted_at

```
CREATE INDEX "idx_community_posts_deleted_at" ON "education"."community_posts"("deleted_at");
```

Index 67: idx_community_posts_post_type

```
CREATE INDEX "idx_community_posts_post_type" ON "education"."community_posts"("post_type");
```

Index 68: idx_community_posts_status

```
CREATE INDEX "idx_community_posts_status" ON "education"."community_posts"("status");
```

Index 69: idx_community_posts_visibility

```
CREATE INDEX "idx_community_posts_visibility" ON "education"."community_posts"("visibility");
```

Index 70: idx_reaction_comment

```
CREATE INDEX "idx_reaction_comment" ON "education"."community_reactions"("comment_id");
```

Index 71: idx_reaction_post

```
CREATE INDEX "idx_reaction_post" ON "education"."community_reactions"("post_id");
```

Index 72: idx_reaction_user

```
CREATE INDEX "idx_reaction_user" ON "education"."community_reactions"("user_id");
```

Index 73: uq_community_tag_name

CREATE UNIQUE INDEX "uq_community_tag_name" ON "education"."community_tags"("name");

Index 74: uq_community_tag_slug

CREATE UNIQUE INDEX "uq_community_tag_slug" ON "education"."community_tags"("slug");

Index 75: idx_community_tags_slug

CREATE INDEX "idx_community_tags_slug" ON "education"."community_tags"("slug");

Index 76: idx_community_tags_usage

CREATE INDEX "idx_community_tags_usage" ON "education"."community_tags"("usage_count" DESC);

9. Baseline Foreign Key DDL from Source SQL

#	Foreign Key Constraint
1	fk_users_role
2	users_domain_role_id_fkey
3	fk_rolepermission_permission
4	fk_rolepermission_role
5	fk_profile_user
6	fk_assessment_user
7	fk_gapreport_user
8	fk_course_educator
9	fk_enrollment_course
10	fk_enrollment_user
11	fk_job_employer
12	fk_application_job
13	fk_application_student
14	fk_notification_user
15	fk_subscription_user
16	fk_lesson_course
17	fk_quiz_lesson
18	fk_assignment_lesson
19	fk_progress_lesson
20	fk_progress_user
21	fk_certificate_course
22	fk_certificate_user
23	fk_achievement_certificate
24	fk_achievement_user
25	fk_task_course
26	fk_task_user
27	fk_recommendation_course
28	fk_recommendation_user
29	fk_announcement_educator
30	fk_auth_otp_user
31	fk_refresh_tokens_user
32	fk_domain_role
33	fk_skill
34	questions_difficulty_id_fkey

35	questions_skill_id_fkey
36	quiz_sessions_domain_role_id_fkey
37	quiz_sessions_user_id_fkey
38	student_answers_difficulty_id_fkey
39	student_answers_question_id_fkey
40	student_answers_session_id_fkey
41	student_answers_skill_id_fkey
42	student_skill_results_session_id_fkey
43	student_skill_results_skill_id_fkey
44	quiz_state_session_id_fkey
45	quiz_state_skill_id_fkey
46	fk_bookmark_post
47	fk_bookmark_user
48	fk_comment_author
49	fk_comment_parent
50	fk_comment_post
51	fk_post_tag_post
52	fk_post_tag_tag
53	fk_community_post_author
54	fk_reaction_comment
55	fk_reaction_post
56	fk_reaction_user

Foreign Key 1: fk_users_role

ALTER TABLE "education"."users" ADD CONSTRAINT "fk_users_role" FOREIGN KEY ("role_id") REFERENCES "education"."roles"("id") ON DELETE NO ACTION ON UPDATE NO ACTION;

Foreign Key 2: users_domain_role_id_fkey

ALTER TABLE "education"."users" ADD CONSTRAINT "users_domain_role_id_fkey" FOREIGN KEY ("domain_role_id") REFERENCES "education"."domain_roles"("domain_role_id") ON DELETE SET NULL ON UPDATE CASCADE;

Foreign Key 3: fk_rolepermission_permission

ALTER TABLE "education"."role_permissions" ADD CONSTRAINT "fk_rolepermission_permission" FOREIGN KEY ("permission_id") REFERENCES "education"."permissions"("id") ON DELETE CASCADE ON UPDATE NO ACTION;

Foreign Key 4: fk_rolepermission_role

ALTER TABLE "education"."role_permissions" ADD CONSTRAINT "fk_rolepermission_role" FOREIGN KEY ("role_id") REFERENCES "education"."roles"("id") ON DELETE CASCADE ON UPDATE NO ACTION;

Foreign Key 5: fk_profile_user

ALTER TABLE "education"."profiles" ADD CONSTRAINT "fk_profile_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;

Foreign Key 6: fk_assessment_user

ALTER TABLE "education"."assessments" ADD CONSTRAINT "fk_assessment_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;

Foreign Key 7: fk_gapreport_user

ALTER TABLE "education"."gap_reports" ADD CONSTRAINT "fk_gapreport_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;

Foreign Key 8: fk_course_educator

ALTER TABLE "education"."courses" ADD CONSTRAINT "fk_course_educator" FOREIGN KEY ("educator_id") REFERENCES "education"."users"("id") ON DELETE SET NULL ON UPDATE NO ACTION;

Foreign Key 9: fk_enrollment_course

ALTER TABLE "education"."enrollments" ADD CONSTRAINT "fk_enrollment_course" FOREIGN KEY ("course_id") REFERENCES "education"."courses"("id") ON DELETE CASCADE ON UPDATE NO ACTION;

Foreign Key 10: fk_enrollment_user

ALTER TABLE "education"."enrollments" ADD CONSTRAINT "fk_enrollment_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;

Foreign Key 11: fk_job_employer

ALTER TABLE "education"."jobs" ADD CONSTRAINT "fk_job_employer" FOREIGN KEY ("employer_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;

Foreign Key 12: fk_application_job

ALTER TABLE "education"."applications" ADD CONSTRAINT "fk_application_job" FOREIGN KEY ("job_id") REFERENCES "education"."jobs"("id") ON DELETE CASCADE ON UPDATE NO ACTION;

Foreign Key 13: fk_application_student

ALTER TABLE "education"."applications" ADD CONSTRAINT "fk_application_student" FOREIGN KEY ("student_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;

Foreign Key 14: fk_notification_user

ALTER TABLE "education"."notifications" ADD CONSTRAINT "fk_notification_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;

Foreign Key 15: fk_subscription_user

ALTER TABLE "education"."subscriptions" ADD CONSTRAINT "fk_subscription_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;

Foreign Key 16: fk_lesson_course

ALTER TABLE "education"."lessons" ADD CONSTRAINT "fk_lesson_course" FOREIGN KEY ("course_id") REFERENCES "education"."courses"("id") ON DELETE CASCADE ON UPDATE NO ACTION;

Foreign Key 17: fk_quiz_lesson

ALTER TABLE "education"."quizzes" ADD CONSTRAINT "fk_quiz_lesson" FOREIGN KEY ("lesson_id") REFERENCES "education"."lessons"("id") ON DELETE CASCADE ON UPDATE NO ACTION;

Foreign Key 18: fk_assignment_lesson

ALTER TABLE "education"."assignments" ADD CONSTRAINT "fk_assignment_lesson" FOREIGN KEY ("lesson_id") REFERENCES "education"."lessons"("id") ON DELETE CASCADE ON UPDATE NO ACTION;

Foreign Key 19: fk_progress_lesson

ALTER TABLE "education"."progress" ADD CONSTRAINT "fk_progress_lesson" FOREIGN KEY ("lesson_id") REFERENCES "education"."lessons"("id") ON DELETE CASCADE ON UPDATE NO ACTION;

Foreign Key 20: fk_progress_user

ALTER TABLE "education"."progress" ADD CONSTRAINT "fk_progress_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;

Foreign Key 21: fk_certificate_course

ALTER TABLE "education"."certificates" ADD CONSTRAINT "fk_certificate_course" FOREIGN KEY ("course_id") REFERENCES "education"."courses"("id") ON DELETE CASCADE ON UPDATE NO ACTION;

Foreign Key 22: fk_certificate_user

ALTER TABLE "education"."certificates" ADD CONSTRAINT "fk_certificate_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;

Foreign Key 23: fk_achievement_certificate

ALTER TABLE "education"."achievements" ADD CONSTRAINT "fk_achievement_certificate" FOREIGN KEY ("certificate_id") REFERENCES "education"."certificates"("id") ON DELETE SET NULL ON UPDATE NO ACTION;

Foreign Key 24: fk_achievement_user

```
ALTER TABLE "education"."achievements" ADD CONSTRAINT "fk_achievement_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 25: fk_task_course

```
ALTER TABLE "education"."tasks" ADD CONSTRAINT "fk_task_course" FOREIGN KEY ("course_id") REFERENCES "education"."courses"("id") ON DELETE SET NULL ON UPDATE NO ACTION;
```

Foreign Key 26: fk_task_user

```
ALTER TABLE "education"."tasks" ADD CONSTRAINT "fk_task_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 27: fk_recommendation_course

```
ALTER TABLE "education"."recommendations" ADD CONSTRAINT "fk_recommendation_course" FOREIGN KEY ("course_id") REFERENCES "education"."courses"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 28: fk_recommendation_user

```
ALTER TABLE "education"."recommendations" ADD CONSTRAINT "fk_recommendation_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 29: fk_announcement_educator

```
ALTER TABLE "education"."announcements" ADD CONSTRAINT "fk_announcement_educator" FOREIGN KEY ("educator_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 30: fk_auth_otps_user

```
ALTER TABLE "education"."auth_otps" ADD CONSTRAINT "fk_auth_otps_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 31: fk_refresh_tokens_user

```
ALTER TABLE "education"."refresh_tokens" ADD CONSTRAINT "fk_refresh_tokens_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 32: fk_domain_role

```
ALTER TABLE "education"."domain_required_skills" ADD CONSTRAINT "fk_domain_role" FOREIGN KEY ("domain_role_id") REFERENCES "education"."domain_roles"("domain_role_id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 33: fk_skill

```
ALTER TABLE "education"."domain_required_skills" ADD CONSTRAINT "fk_skill" FOREIGN KEY ("skill_id") REFERENCES "education"."skills"("skill_id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 34: questions_difficulty_id_fkey

```
ALTER TABLE "education"."questions" ADD CONSTRAINT "questions_difficulty_id_fkey" FOREIGN KEY ("difficulty_id") REFERENCES "education"."difficulty_levels"("difficulty_id") ON DELETE NO ACTION ON UPDATE NO ACTION;
```

Foreign Key 35: questions_skill_id_fkey

```
ALTER TABLE "education"."questions" ADD CONSTRAINT "questions_skill_id_fkey" FOREIGN KEY ("skill_id") REFERENCES "education"."skills"("skill_id") ON DELETE NO ACTION ON UPDATE NO ACTION;
```

Foreign Key 36: quiz_sessions_domain_role_id_fkey

```
ALTER TABLE "education"."quiz_sessions" ADD CONSTRAINT "quiz_sessions_domain_role_id_fkey" FOREIGN KEY ("domain_role_id") REFERENCES "education"."domain_roles"("domain_role_id") ON DELETE RESTRICT ON UPDATE NO ACTION;
```

Foreign Key 37: quiz_sessions_user_id_fkey

```
ALTER TABLE "education"."quiz_sessions" ADD CONSTRAINT "quiz_sessions_user_id_fkey" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 38: student_answers_difficulty_id_fkey

```
ALTER TABLE "education"."student_answers" ADD CONSTRAINT "student_answers_difficulty_id_fkey" FOREIGN KEY ("difficulty_id") REFERENCES "education"."difficulty_levels"("difficulty_id") ON DELETE RESTRICT ON UPDATE NO ACTION;
```

Foreign Key 39: student_answers_question_id_fkey

```
ALTER TABLE "education"."student_answers" ADD CONSTRAINT "student_answers_question_id_fkey" FOREIGN KEY ("question_id") REFERENCES "education"."questions"("question_id") ON DELETE RESTRICT ON UPDATE NO ACTION;
```

Foreign Key 40: student_answers_session_id_fkey

```
ALTER TABLE "education"."student_answers" ADD CONSTRAINT "student_answers_session_id_fkey" FOREIGN KEY ("session_id") REFERENCES "education"."quiz_sessions"("session_id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 41: student_answers_skill_id_fkey

```
ALTER TABLE "education"."student_answers" ADD CONSTRAINT "student_answers_skill_id_fkey" FOREIGN KEY ("skill_id") REFERENCES "education"."skills"("skill_id") ON DELETE RESTRICT ON UPDATE NO ACTION;
```

Foreign Key 42: student_skill_results_session_id_fkey

```
ALTER TABLE "education"."student_skill_results" ADD CONSTRAINT "student_skill_results_session_id_fkey" FOREIGN KEY ("session_id") REFERENCES "education"."quiz_sessions"("session_id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 43: student_skill_results_skill_id_fkey

```
ALTER TABLE "education"."student_skill_results" ADD CONSTRAINT "student_skill_results_skill_id_fkey" FOREIGN KEY ("skill_id") REFERENCES "education"."skills"("skill_id") ON DELETE RESTRICT ON UPDATE NO ACTION;
```

Foreign Key 44: quiz_state_session_id_fkey

```
ALTER TABLE "education"."quiz_state" ADD CONSTRAINT "quiz_state_session_id_fkey" FOREIGN KEY ("session_id") REFERENCES "education"."quiz_sessions"("session_id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 45: quiz_state_skill_id_fkey

```
ALTER TABLE "education"."quiz_state" ADD CONSTRAINT "quiz_state_skill_id_fkey" FOREIGN KEY ("skill_id") REFERENCES "education"."skills"("skill_id") ON DELETE NO ACTION ON UPDATE NO ACTION;
```

Foreign Key 46: fk_bookmark_post

```
ALTER TABLE "education"."community_bookmarks" ADD CONSTRAINT "fk_bookmark_post" FOREIGN KEY ("post_id") REFERENCES "education"."community_posts"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 47: fk_bookmark_user

```
ALTER TABLE "education"."community_bookmarks" ADD CONSTRAINT "fk_bookmark_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 48: fk_comment_author

```
ALTER TABLE "education"."community_comments" ADD CONSTRAINT "fk_comment_author" FOREIGN KEY ("author_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 49: fk_comment_parent

```
ALTER TABLE "education"."community_comments" ADD CONSTRAINT "fk_comment_parent" FOREIGN KEY ("parent_id") REFERENCES "education"."community_comments"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 50: fk_comment_post

```
ALTER TABLE "education"."community_comments" ADD CONSTRAINT "fk_comment_post" FOREIGN KEY ("post_id") REFERENCES "education"."community_posts"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 51: fk_post_tag_post

```
ALTER TABLE "education"."community_post_tags" ADD CONSTRAINT "fk_post_tag_post" FOREIGN KEY ("post_id") REFERENCES "education"."community_posts"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 52: fk_post_tag_tag

```
ALTER TABLE "education"."community_post_tags" ADD CONSTRAINT "fk_post_tag_tag" FOREIGN KEY ("tag_id") REFERENCES "education"."community_tags"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 53: fk_community_post_author

```
ALTER TABLE "education"."community_posts" ADD CONSTRAINT "fk_community_post_author" FOREIGN KEY ("author_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 54: fk_reaction_comment

```
ALTER TABLE "education"."community_reactions" ADD CONSTRAINT "fk_reaction_comment" FOREIGN KEY ("comment_id") REFERENCES "education"."community_comments"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 55: fk_reaction_post

```
ALTER TABLE "education"."community_reactions" ADD CONSTRAINT "fk_reaction_post" FOREIGN KEY ("post_id") REFERENCES "education"."community_posts"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

Foreign Key 56: fk_reaction_user

```
ALTER TABLE "education"."community_reactions" ADD CONSTRAINT "fk_reaction_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
```

10. Functions and Triggers from Source SQL

No CREATE FUNCTION statements were found in the provided Edu_saas_DB_Scripts.sql source.

No CREATE TRIGGER statements were found in the provided Edu_saas_DB_Scripts.sql source.

11. Added AI Support Indexes

```
CREATE INDEX IF NOT EXISTS idx_activity_logs_user_id ON education.activity_logs(user_id);
CREATE INDEX IF NOT EXISTS idx_activity_logs_last_activity ON education.activity_logs(last_activity);
CREATE INDEX IF NOT EXISTS idx_activity_logs_user_last_activity ON education.activity_logs(user_id,last_activity);
CREATE INDEX IF NOT EXISTS idx_login_history_user_id ON education.login_history(user_id);
CREATE INDEX IF NOT EXISTS idx_login_history_days_since_last_login ON education.login_history(days_since_last_login);
CREATE INDEX IF NOT EXISTS idx_course_skills_course_id ON education.course_skills(course_id);
CREATE INDEX IF NOT EXISTS idx_course_skills_skill_id ON education.course_skills(skill_id);
CREATE INDEX IF NOT EXISTS idx_course_prerequisites_course_id ON education.course_prerequisites(course_id);
CREATE INDEX IF NOT EXISTS idx_course_prerequisites_prerequisite_course_id ON education.course_prerequisites(prerequisite_course_id);
CREATE INDEX IF NOT EXISTS idx_course_ratings_user_id ON education.course_ratings(user_id);
CREATE INDEX IF NOT EXISTS idx_course_ratings_course_id ON education.course_ratings(course_id);
CREATE INDEX IF NOT EXISTS idx_course_ratings_rating ON education.course_ratings(rating);
CREATE INDEX IF NOT EXISTS idx_recommendations_rank ON education.recommendations(rank);
CREATE INDEX IF NOT EXISTS idx_recommendations_confidence_score ON education.recommendations(confidence_score);
CREATE INDEX IF NOT EXISTS idx_recommendations_user_rank ON education.recommendations(user_id, rank);
```

12. Added AI Support Function and Triggers

```
CREATE OR REPLACE FUNCTION education.set_updated_at()
RETURNS TRIGGER AS $$
BEGIN
```

```

        NEW.updated_at = CURRENT_TIMESTAMP;
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_activity_logs_updated_at
BEFORE UPDATE ON education.activity_logs
FOR EACH ROW EXECUTE FUNCTION education.set_updated_at();

CREATE TRIGGER trg_login_history_updated_at
BEFORE UPDATE ON education.login_history
FOR EACH ROW EXECUTE FUNCTION education.set_updated_at();

CREATE TRIGGER trg_course_skills_updated_at
BEFORE UPDATE ON education.course_skills
FOR EACH ROW EXECUTE FUNCTION education.set_updated_at();

CREATE TRIGGER trg_course_prerequisites_updated_at
BEFORE UPDATE ON education.course_prerequisites
FOR EACH ROW EXECUTE FUNCTION education.set_updated_at();

CREATE TRIGGER trg_course_ratings_updated_at
BEFORE UPDATE ON education.course_ratings
FOR EACH ROW EXECUTE FUNCTION education.set_updated_at();

CREATE TRIGGER trg_recommendations_updated_at
BEFORE UPDATE ON education.recommendations
FOR EACH ROW EXECUTE FUNCTION education.set_updated_at();

```

CONFIDENTIAL
VAYVORA™ TECHNOLOGIES PVT. LTD.
PROPRIETARY INFORMATION
DO NOT DISTRIBUTE

13. Other Database Objects

Object Type	Count	Notes
Views	0	No view DDL found in source SQL
Sequences	0	No explicit CREATE SEQUENCE found; generated implicitly for SERIAL columns
Seed data	0	No INSERT statements found in source SQL
Comments	0	No COMMENT ON statements found in source SQL
Grants	0	No GRANT statements found in source SQL

14. Catalog Completion Notes

This catalog includes every SQL object category found in the provided Edu_saas_DB_Scripts.sql file plus the approved AI support additions from the working design. The source SQL does not contain function, trigger, view, sequence, seed data, comment, or grant statements. The only function/trigger DDL in this catalog is therefore the explicitly added AI-support updated_at function and triggers.